MIL-F-51215C(EA)
27 October 1980
SUPERSEDING
MIL-F-51215B(EA)
19 November 1974

MILITARY SPECIFICATION

FILTERS, PARTICULATE, 600, 1200, 2500 AND 5000 CFM; C18R1, C19R1, C30R1 AND C20R1

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of particulate filters.

1.2 Types. The particulate filters shall be furnished in the following types as specified (see 6.2).

Type	Capacity-cubic feet per minute (CFM) (Capacity meters per minute (Cu m/per min
C18R1	600 (16.992)
C19R1	1200 (33.984)
C30R1	2500 (70.8)
C20R1	5000 (141.6)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 4240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-F-51079 - Filter Medium, Fire-Resistant, High Efficiency.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-282 - Filter Units, Protective Clothing, Gas Mask Components and Related Products.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

D5-19-1862 - Filter, Particulate, 600 cfm, C18R1.
D5-19-1867 - Filter, Particulate, 1200 cfm, C19R1.
D5-19-1853 - Filter, Particulate, 2500 cfm, C30R1.
E5-19-1848 - Filter, Particulate, 5000 cfm, C20R1.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

E84 - Surface Burning Characteristic of Building Materials.

(Copies of this standard may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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UNDERWRITERS LABORATORIES, INC.

UL Standard

UL 723 - Test Method for Fire Hazard Classification of Building Materials.

(Copies of this standard may be obtained from the Underwriters Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on the applicable data list shall conform to the specifications and subsidiary drawings listed thereon or to the specific characteristics set forth on the drawings.

3.1.1.1 Plywood. The plywood used in the construction of the filters shall have a flame spread classification of 25 or less when tested as specified in ASTM Standard E84 or UL723.

3.1.2 Components. All components for the respective particulate filters shall conform to the specifications and drawings listed on the applicable data list.

3.2 Manufacture and assembly. The filters shall be assembled and marked in accordance with the following drawings:

C18R1 - D5-19-1862
C19R1 - D5-19-1867
C30R1 - D5-19-1853
C20R1 - E5-19-1848

The faces of the filter shall be flat and parallel to within a total allowance of 1/16 (1.587 mm) inch. The filter shall be square to within a total allowance of 1/8 (3.175 mm) inch when measured diagonally across the corners of both faces through which air flows.

3.3 Airflow resistance and dioctylphthalate (DOP) penetration. The particulate filters shall conform to the airflow resistance and total DOP penetration (through the filter medium frame and gasket which has been encapsulated) requirements listed in table I when tested as specified in 4.4.4.1. Unless otherwise specified, the filters shall be tested at a flow rate specified in table I and at 20 percent of the rated airflow for all size filters except for the C20R1.

3.4 Resistance to rough handling. After rough handling as specified in 4.4.4.2 for 15 minutes at 3/4 inch (19.05 mm) amplitude with the pleats, faces

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TABLE I. Airflow resistance and DOP penetration

Property	Type			
	C18R1	C19R1	C30R1	C20R1
Airflow resistance, inches of water (max)	1.25 (31.75 mm)	1.25 (31.75 mm)	0.50 (12.7 mm)	0.20 (5.08 mm)
DOP penetration, percent (max)	0.03	0.03	0.02	0.01
Test air flow rate (cfm)	600	1000	1000	1000
20 percent of rated flow	120	240	500	----

and separators in the vertical position, the filters shall comply with the airflow resistance and DOP penetration requirements of 3.3.

3.5 Marking. Each filter shall be marked by stencilling or with a non-removable label on the top of the frame to indicate the manufacturer, model, serial number, test airflow direction, airflow resistance at rated airflow and the results of the dioctylphthalate (DOP) smoke penetration test at rated airflow and at 20 percent of rated airflow for all sizes of filters. In addition the wording "Install With This Side Up" shall be stencilled on the top of the frame in 3/8 inch minimum height letters. The top of the filter shall be perpendicular to the pleats in the filter medium.

3.6 Preproduction. Prior to the start of regular production, a preproduction lot of particulate filters shall be produced in accordance with this specification for examination and tests (see 4.3).

3.7 Workmanship. The filters shall be free from foreign matter (dirt, grease, oily or viscous material) and damage, such as distorted or cracked frame, deformation or sagging of medium, separators, and faceguards, cracks in adhesive, and cracks and holes in exposed portions of the medium.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of six each of the types of filters specified in the contractor order shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production (see 6.2).

4.3.2 Inspection procedure.

4.3.2.1 For examination. Each filter in the preproduction lot shall be examined for all requirements of the applicable drawings and this specification.

4.3.2.2 For tests. Each filter in the preproduction lot shall be tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. Each filter in the preproduction lot shall comply with the examination and tests specified in 4.3.2 to be acceptable.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the completed filter of the same size produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions. However, no more than one lot of filter medium in one week's filter production and no more than 250 filters shall be represented in any one lot of assembled filters.

4.4.2 Sampling for examination and test. Sampling for examination and test shall be conducted in accordance with MIL-STD-105. For Major defect 101, resistance to rough handling, level S-4 shall be used.

4.4.3 Inspection procedure.

4.4.3.1 For examination and test. Sample filters shall be examined and tested in accordance with the classification of defects and with MIL-STD-105. The sample filters (Major 101) shall be tested in accordance with 4.4.4.1 after testing in accordance with 4.4.4.2.

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4.4.3.2 For critical defects. Each filter in the lot shall be tested for it's critical characteristics listed in the classification of defects.

4.4.3.3 Classification of defects.

(a) Filters, particulate, 600, 1200, 2500, 5000 cfm (Dwg D5-19-1862, D5-19-1867, D5-19-1853, and E5-19-1848).

Categories	Defects	Acceptance standards
Critical:		
1	Airflow resistance	4.4.4.1
2	DOP smoke penetration	4.4.4.1
Major:	AQL 2.5 percent defective	
101	Resistance to rough handling	4.4.4.2
	AQL 1.0 percent defective	
102	Component missing	
103	Component incorrectly assembled	
104	Marking incorrect	
105	Location of strikes incorrect	
106	Flatness and squareness (3.2)	
107	Workmanship (3.7)	
Minor:	AQL 2.5 percent defective	
201	Marking illegible	

(b) Packaging.

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Filter inlet and outlet openings not covered as specified
102	Loose pack
103	Box closure incorrect
104	Strapping loose
105	Marking incorrect or illegible

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4.4.4 Tests.

4.4.4.1 Airflow resistance and DOP smoke penetration. The filter shall be encapsulated and tested for airflow resistance and DOP smoke penetration through the filter medium, frame and gasket using DOP Filter Testing Penetrometer, Q107.

4.4.4.2 Resistance to rough handling. The filters shall be rough handled in accordance with Test Method T-105.10 of MIL-STD-282. They shall be examined for damage and the airflow resistance and DOP smoke penetration determined in accordance with 4.4.4.1. Filters shall be tested without packing or packaging.

5. PACKAGING

5.1 Preservation and unit packing.

5.1.1 Level A. Each filter shall be preserved and unit packed in accordance with 5.4.

5.2.1 Level A and commercial. Each filter, unit packed as specified in 5.1, shall be packed in accordance with 5.4.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.3 Marking. In addition to any special marking required, the contract or order, unit packs and shipping containers shall be marked in accordance with 5.4.

5.4 Repair part. When this item is procured for storage and issue as a repair part, preservation, unit and intermediate packing, packing, and marking shall be as specified on the packaging data sheet which is identified by the item National Stock Number, and/or part number.

6. NOTES

6.1 Intended use. The particulate filters covered by this specification are components of filter units for protecting fixed installations against toxic gases and particulate matter. The following is a list of the particulate filters and their respective assemblies:

Particulate Filter Type	Assembly
C18R1	Filter, Gas-Particulate, 600 cfm, M14A1.
C19R1	Filter, Gas-Particulate, 1200 cfm, M15A1.
C30R1	Filter, Gas-Particulate, 2500 cfm, M16A1.
C20R1	Filter, Gas-Particulate, 5000 cfm, M17A1.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of particulate filter required (see 1.2).
(c) Level of packing required (see 5.2).
(d) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether to proceed with production.

6.3 Manufacturing procedure. Filters manufactured in the following manner have been found to satisfy the performance requirements of this specification.

(a) Filter material conforming to MIL-F-51079 should be pleated on a suitable pleating machine using fiberboard separators to form and hold the pleats. A section of pleated material should be cut off containing the number of folds which has been found to give the desired compactness. This section should be placed on a temporary U frame of approximately the same dimensions as the filter being assembled. The exposed fiberboard separators should be removed and fluted aluminum separators, which are approximately the same length as the filter, inserted in their place. Any torn places in the filter material should be repaired with adhesive for filters. The pleated assembly should be turned over and the operation repeated.

(b) The pleated section should be checked for proper compactness. Excessively tight packing will cause the fluted separators to cut into the filter material, and excessively loose packing will allow undesirable movement of the separators. Correct compactness may be obtained by removing or splicing in sections of pleated filter material to the ends of the assembly.

(c) Fiberboard backing strips should be inserted between each end of the pleated assembly and the temporary frame to keep pleats from bending over when trimmed. Both sides of the pleated assembly should be trimmed with a high speed band saw to obtain correct width. It is necessary to have an angle of no less than 30 degrees between the saw blade and filter pleats when trimming to prevent the saw from tearing the paper.

(d) One coat of adhesive for filters should be applied to both sides of the pleated assembly and allowed to penetrate and dry approximately 2 hours. A second coat of adhesive should be applied and allowed to penetrate and dry approximately 4 hours. A third coat of adhesive should be applied and allowed to penetrate and dry approximately 4 hours. Additional coats should be applied if necessary to cover any pin holes which might arise when solvent evaporates. After careful inspection to see that there are no pin holes remaining, the

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excess adhesive should be trimmed from the edges of the assembly and the temporary frame removed.

(e) The inner and outer panels should be assembled around the pleated assembly to form a permanent frame. The pleated filter assembly should be centered to allow approximately 1/8 inch of the frame to extend on either side of the insertion of the faceguards inside the frame. One coat of adhesive should be applied around the entire periphery, sealing the frame to the filter. The adhesive should be allowed to penetrate and dry approximately 2 hours. A second coat of adhesive should then be applied and allowed to penetrate and dry approximately 4 hours. The filter should then be turned over and the cementing operation repeated on the reverse side. The plywood or fiberboard cover should be fastened to the flush side of the filter to prevent damage during further handling. All holes in the edges of the frame should be filled with felt and the edges coated with adhesive.

(f) The final test for DOP penetration and air resistance shall be performed and the filter marked in accordance with the applicable drawing.

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 214. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization officer, if required.

Custodian:

Army - EA

Preparing activity:

Army - EA

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